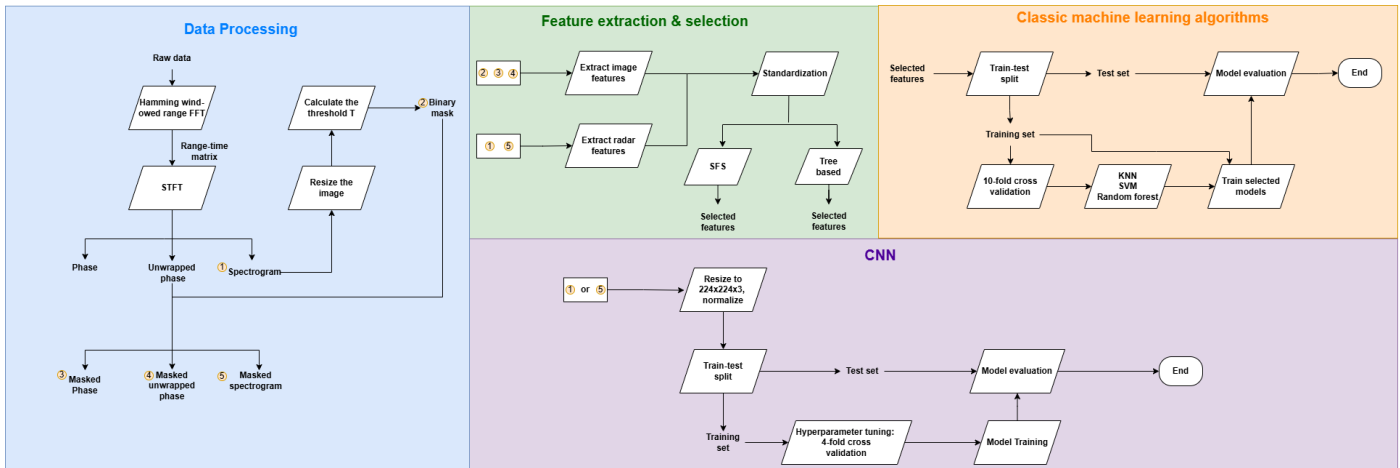


EE 4675 Object Classification with Radar Project Report

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This page describes the workflow of the course EE 4675 project. The aim of this project is to classify six activities of human subjects.

From experiments, radar I/Q data are collected and referred to as raw data in the flowchart. With windowing and Fourier transform, the range-time matrices are obtained. Following the Short Time Fourier Transform (STFT), spectrogram, phase, and unwrapped phase images are generated. By combining both the amplitude and phase information, more distinctive features can be discovered. Then the spectrogram is binarized to obtain the mask, with the threshold selected based on the mean value inside the spectrogram. The mask is further applied to three types of images, to produce the masked spectrogram, masked phase, and masked unwrapped phase images, correspondingly. The mask allows us to focus only on the Region of Interest (ROI) containing the contribution of the moving targets. Areas that do not convey salient information, such as the portion with low energy should be discarded.

For feature extraction and classification, two separate routes are selected: classical machine learning algorithms and Convolutional Neural Network (CNN).

In the first route, 68 ROI and textural image features (perimeter, area, local binary pattern, etc) and 21 radar features (centroid, bandwidth, SVD, etc) are extracted for each sample. To reduce the time and storage in training, Sequential Feature Selection (SFS) and Extra Trees Classifier are used for feature selection. After splitting the training set (80%) and test set (20%), features standardization is performed. Next, 10-fold cross validation is carried to tune hyperparameters and select the best model. Three models are considered, namely K-Nearest Neighbors (KNN), Support Vector Machine (SVM) and Random Forest. The best accuracy on test set is **91.10%** with Random Forest.

In the second route with CNN, two different inputs are used: RGB spectrogram and masked spectrogram, and two different models are considered: ConvNet and ConvSE. The ConvNet is a simplified version of VGG13, with 6 convolution layers and 3 linear layers. ConvSE enhances this by concatenating feature maps from deep and shallow levels and using a Squeeze-and-Excitation (SE) Module to adjust the importance of each feature map. The SE Module recalibrates channel-wise feature responses, allowing the network to focus dynamically on the most informative channels. All models use the Focal Loss function, which emphasizes confusable samples. After hyperparameter tuning and training, the model is evaluated on accuracy and confusion matrix. The ConvSE model with spectrogram input achieves the highest test accuracy: **93.56%**.

Based on the classification results, we find the “picking up an object” and “drinking water” activities are the most difficult to distinguish because of the intrinsic similarity. This error is minimized by designing better features and models in this project.